

Simple Mohr's Circle Computation Questions Typical for Exam 4

1. Given I_{AVG} and radius (or diameter), determine I_{MAX} and/or I_{MIN} .

(The problem may say that the "center" is at some value. Realize that the "center" is I_{AVG} .)

Question 1

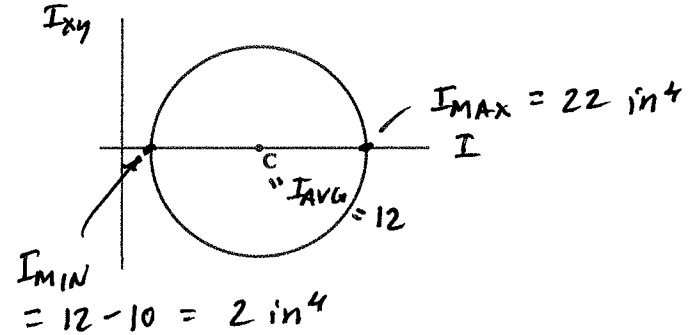
Given I_{AVG} and radius, find I_{MAX} and I_{MIN} .

Center $I_{AVG} =$	12.00	in^4
Radius $R =$	10.00	in^4

Solution:

$$I_{AVG} = \text{center of MC} = \frac{I_x + I_y}{2}$$

Picture (usually not provided on the exam, but provided here to help you visualize the solution)



2. Given I_{AVG} and diameter, determine I_{MAX} and/or I_{MIN} .

(The problem may say that the "center" is at some value. Realize that the "center" is I_{AVG} .)

Question 2:

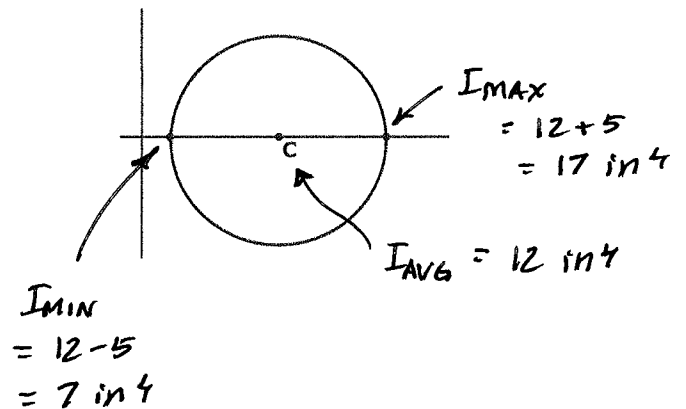
Given I_{AVG} and diameter, find I_{MAX} and I_{MIN} .

Center $I_{AVG} =$	12.00	in^4
Diameter $d =$	10.00	in^4

Solution:

$$\begin{aligned} \text{dia} &= 10 \text{ in}^4 \\ R &= \frac{\text{dia}}{2} = 5 \text{ in}^4 \end{aligned}$$

Picture (usually not provided on the exam, but provided here to help you visualize the solution)



3. Given I_{AVG} and point B (I_B, I_{BD}) (note that point B is sometimes called a "reference point on the circle), determine the following:

Radius (R), I_{MAX} , I_{MIN} , point D (90 degrees from point B on the area), Angles to principal axes θ_{p1} , θ_{p2} .

(The problem may say that the "center" is at some value. Realize that the "center" is I_{AVG} .)

Question 3:

Given I_{AVG} and point B [I_B, I_{BD}]:

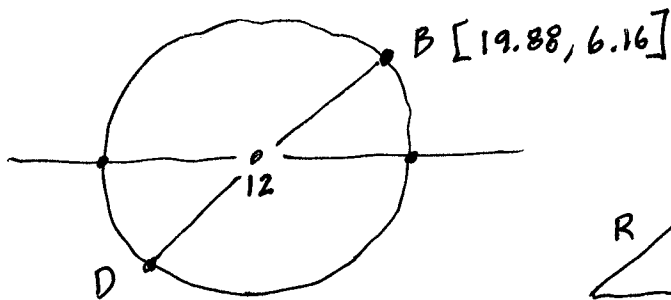
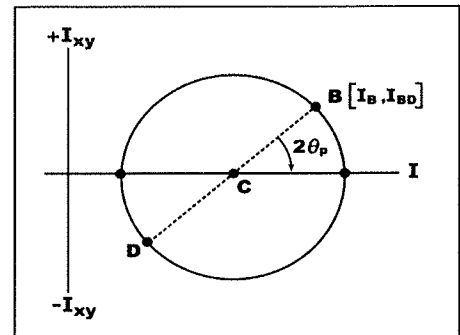
Center I_{AVG} =	12.00	in^4
Coordinates of point B [I_B, I_{BD}]:	I_B =	19.88 in^4
	I_{BD} =	6.16 in^4

Please determine:

Radius R, I_{MAX} , I_{MIN} , point D [I_D, I_{BD}] (90 deg from point B on area), θ_{p1} , θ_{p2}

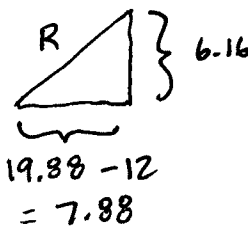
Solution:

Picture (usually not provided on the exam, but provided here to help you visualize the solution)



Since $R = 10 \text{ in}^4$,

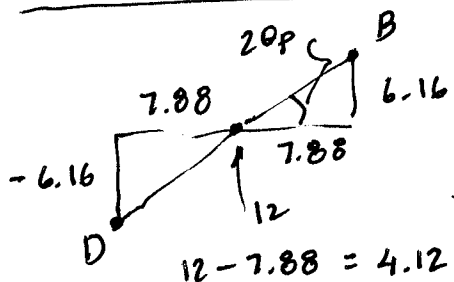
$$\begin{aligned} I_{MAX} &= 12 + 10 = 22 \text{ in}^4 \\ I_{MIN} &= 12 - 10 = 2 \text{ in}^4 \end{aligned}$$



$$\begin{aligned} R &= \sqrt{7.88^2 + 6.16^2} \\ &= \text{norm}([7.88, 6.16]) \end{aligned}$$

$$R = 10.0 \text{ in}^4$$

Coords of D = Mirror Image of B :



$$\text{Coords of D} = [4.12, -6.16] \text{ in}^4$$

$$2\theta_p = \tan^{-1} \frac{6.16}{7.88}$$

$$= 38.0^\circ$$

$$\theta_{p1} = 19.0^\circ$$

$$\theta_{p2} = 19 + 90 = 109^\circ$$

4. Given I_{AVG} , Radius (R), and angle ϕ CCW from I_{MAX} , determine the following:
 I_{MAX} , I_{MIN} , and point B.

Question 4

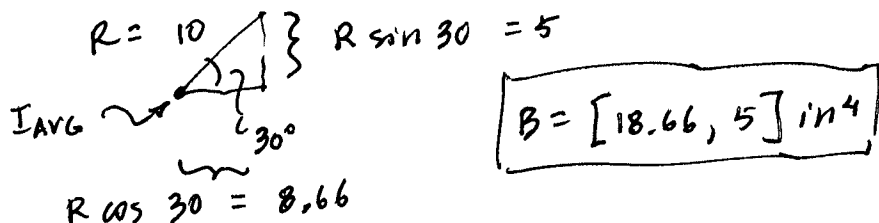
Given I_{AVG} , Radius and angle ϕ CCW from I_{MAX} :

Center I_{AVG} =	12.00	in^4
Radius R =	10.00	in^4
ϕ =	15.0	degrees (= 20 on MC)

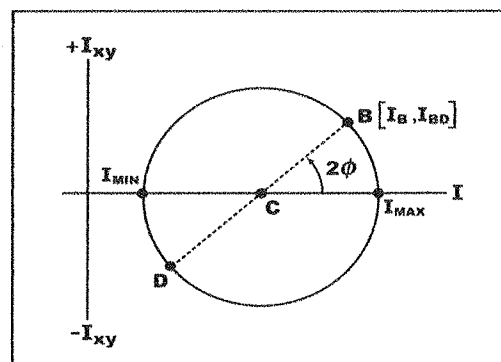
(Point B in upper right quadrant of MC)

Please determine: I_{MAX} , I_{MIN} , point B [I_B, I_{BD}] (at ϕ deg CCW from I_{MAX})

Solution: $I_{AVG} = 12 \text{ in}^4$



Picture (usually not provided on the exam, but provided here to help you visualize the solution)



5. Given I_{AVG} , Radius (R), and angle ϕ CW from I_{MIN} , determine the following:
 I_{MAX} , I_{MIN} , and point B.

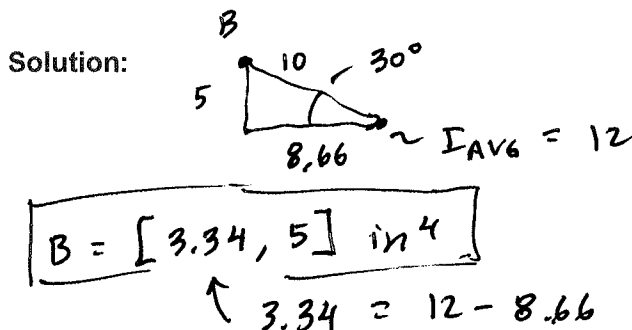
Question 5:

Given I_{AVG} , Radius and angle ϕ CW from I_{MIN} :

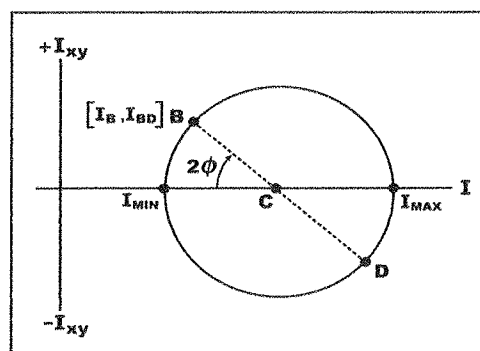
Center I_{AVG} =	12.00	in^4
Radius R =	10.00	in^4
ϕ =	15.0	degrees (= 20 on MC)

(Point B in upper left quadrant of MC)

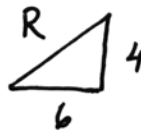
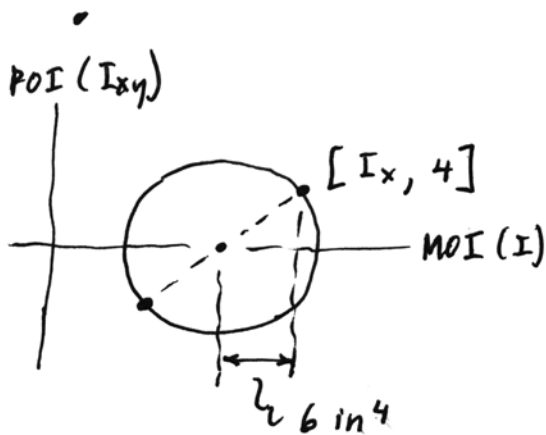
Please determine: I_{MAX} , I_{MIN} , point B [I_B, I_{BD}] (at ϕ deg CW from I_{MIN})



Picture (usually not provided on the exam, but provided here to help you visualize the solution)



6. Given: I at a particular point is 6 in^4 larger or smaller than the center. POI at that point is 4 in^4 . Find R .



$$R = \sqrt{6^2 + 4^2}$$

$$R = 7.21 \text{ in}^4$$

7. Example Problem

Two axes $P + Q$,

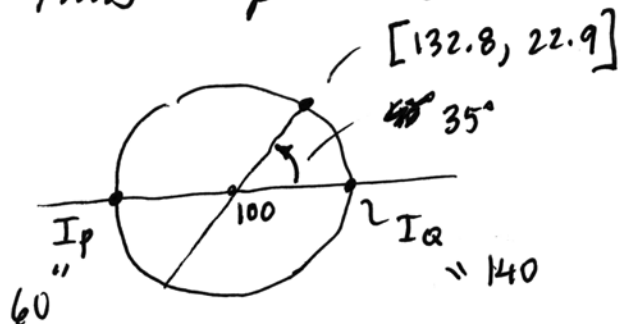
$$I_P = 60 \text{ in}^4$$

$$I_Q = 140 \text{ in}^4$$

Area is symmetric about axis P ,

This tells us that $I_{xy} = 0$.

Thus $I_P + I_Q$ are principals.



Find I_{xy} on MC
WRT axes rotated
at 35° on MC ,
CCW

$$R = \frac{140 - 60}{2} = 40$$

